

CONFIRMED LIVE · 17:43 PT / 15:43 UTC · 18 August 2026

JFTSE SeaWave LTR — first working cave

(2026-08-18)

Addendum to the morning origin-findings note on branch test/wave-origin-directions (ThewindMom/JFTSE). That note said left-to-right travel was impossible. It was wrong about the travel axis. This document is the confirmed first LTR cave and the spawn that matches tonight's tape.

Status — Thewind confirmed this look again tonight

17:43 PT (15:43 UTC) 18 August 2026. Foam is a vertical band that travels sideline ↔ sideline (left → right). The working client is the first cave (mov ecx, edx / 8B CA), not the later Z-spread-zero cave. Spawn is five waves, 5 s apart, attacker=4, xyz = (−200, 0, 0) “net center, further left”.

What this is / is not

- Hand-off note for another JFTSE person. Bytes, hashes, and script values below were read from the live files on this box at write time.
- This is **not** a git commit. Do not commit or push from this note. The existing branch already has the earlier -wavetest harness and the origin-findings PDF.
- Times are Europe/Berlin, tagged PT. UTC is in parentheses when it matters.
- Anything not re-read from a file or not confirmed by Thewind is marked **unknown**.

1. Court coordinates (confirmed live)

UseSkill xyz plants the **drop only**. There is no heading field in the packet. Constructor is S2CMatchplayUseSkill(attacker, target, skillId, seed, xTarget, zTarget, yTarget). Chat labels print that as xyz=(x, z, y). For the confirmed spawn all three numbers after x are zero, so the order does not change the value.

(0, 0, 0)	Net-center. Not under the player.
−X	Left sideline. +X = right sideline.
+Y	Far / opponent baseline.
−Y	Player / near baseline.
Packet floats	Order is xTarget, zTarget, yTarget. zTarget (2nd) is height. yTarget (3rd) is court length / client Z. Confirmed spawn uses 0, 0 after x.
Heading	None. Packet does not carry a travel yaw.

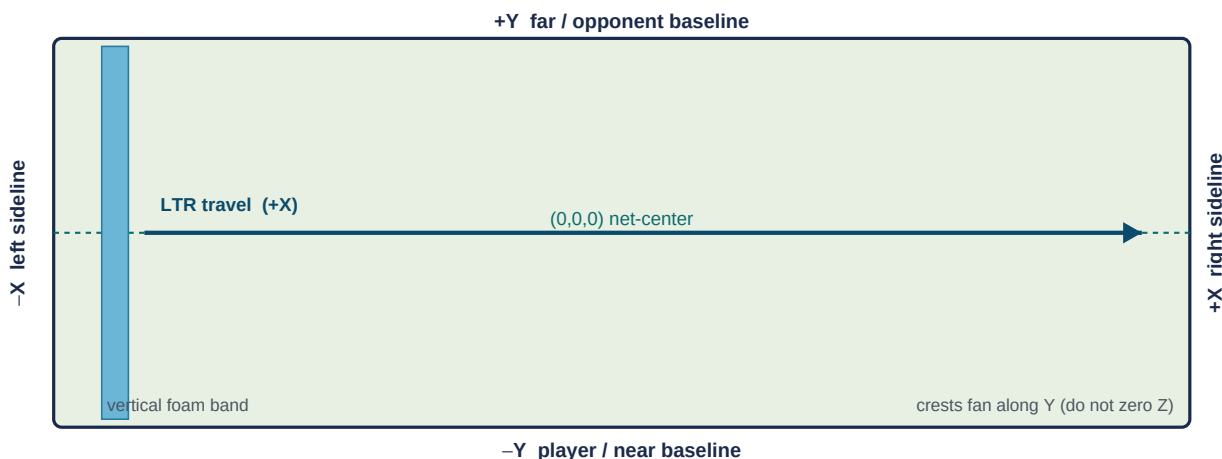


Diagram is schematic (not to scale). The blue strip is the foam band; the arrow is travel. Vanilla SeaWave is the opposite: a full-width band along X that travels along Y.

Sideline scale (client X) is about ± 55 . Morning drops at ± 150 were already at the visible sideline. Confirmed spawn $x = -200$ is already past the left edge, so $x = -500$ does not look further left (saturation, not a swapped axis). Left-right is still `step.x`. `step.y` walks baseline $\leftrightarrow$ net.

2. Vanilla SeaWave (unpatched client)

From the 18 August client RE note (evidence/seawave-ltr/README.md) and the morning origin-findings PDF. Skill identity was not re-disassembled for this addendum; the live patch still keys on skillId 27.

Field	Value	Notes
Packet skillId	27	Same id -wavetest fires. No new skill.
Skill table id	28	From origin-findings / RE note.
ShotType	6	Mover at VA 0x4E9CE0 handles types 3/6/7.
Targeting	13	UseSkill xyz $\rightarrow$ instance+0x4C/+0x50/+0x54. No heading float.
ShotCnt	11	Eleven crests. Hardcoded for skillId 27.
ShotRot	0	Turn-rate in deg/s, not yaw. 0 $\Rightarrow$ dir never steers.

First-frame direction (skillId 27 only)

At the first-frame dir build (`cmp [ebp+8], 0x1B` at VA 0x4E9D91) the client sets, for shot index 0..10:

```
dir.x = hardcoded spread -0.5, -0.4, ..., 0, ..., +0.5 (11 crests)
dir.y = FPU leftover (~0)
dir.z = attacker-related Z
then D3DXVec3Normalize into projectile+0x54 / +0x58 / +0x5C
later ticks freeze that dir and integrate pos += dir * speed * dt
```

Normalize of (small X, 0, large Z) is why vanilla foam is a **full-width wall traveling baseline $\leftrightarrow$ net**. A FieldItem / XML clone cannot go sideways: the 11-piece spread and this dir build are hardcoded to skillId 27. TPosition is unused for ShotType 6. ShotRot = 90 only enables steering toward dest; it does not rotate the wall. A new skill id loses the 11-crest spread.

3. What “LTR” means (user-corrected)

Early analysis that said “still baseline $\leftrightarrow$ net” was wrong

Thewind at 15:36 PT: “i see it going left to right though.” Foam is a **vertical band** that travels **sideline $\leftrightarrow$ sideline** (left-right). A wide/tall wall is not the same as the travel axis. The morning origin-findings PDF (“Left-to-right travel is not possible”) is superseded for travel. xyz still only moves the drop. That part stands.

	Vanilla (unpatched)	LTR (this cave)
Foam shape	Horizontal band (sideline $\leftrightarrow$ sideline already)	Vertical band (baseline $\leftrightarrow$ baseline already)
Travel axis	Along Y — baseline $\leftrightarrow$ net	Along +X — left $\rightarrow$ right (sideline $\leftrightarrow$ sideline)
11 crests	Spread in X (width of the wall)	Old X-spread moved onto Z: crests fan along the court
Drop xyz	Plants spawn only	Same. Does not aim the wave.

RTL (right $\rightarrow$ left) would be the same cave with BA 00 00 80 BF (−1.0) instead of BA 00 00 80 3F (+1.0). RTL was not the confirmed take.

4. Working exe patch — FIRST LTR cave

This is the cave Thewind confirmed again tonight. Image base 0x400000. File offset = VA − 0x400000 for .text. Verified on live /workspace/jftse-work/client/FantaTennis.exe at write time.

Item	Value
Site VA / file	VA 0x4EA115 / file 0x0EA115
Site size	14 bytes $\rightarrow$ jmp 0x6E5C58 + 9 nops

Item	Value
Site bytes (live)	E9 3E BB 1F 00 90 90 90 90 90 90 90 90 90 90
Cave VA / file	VA 0x6E5C58 / file 0x2E5C58
Cave size	30 bytes in .text raw padding
.text VirtualSize	was 0x2E4C58 → now 0x2E4C80 (file 0x220)
SectionAlignment	0x1000 already maps this page. SizeOfImage unchanged (from RE note).
Return	jmp 0x4EA125 (existing normalize + integrate)
Index-fail path	0x4EA123 dd d8 (fstp st0) is not touched

Cave logic (skillId 27 only)

```

cmp [ebp+8], 0x1B      ; skillId 27
jne +7                ; other skills: original edx/eax/ecx stores
mov ecx, edx           ; 8B CA dir.z = old dir.x (-0.5 ... +0.5 fan)
mov edx, 0x3F800000    ; BA 00 00 80 3F dir.x = +1.0 (LTR)
; then the original three stores, then jmp 0x4EA125
mov [esp+0x18], edx    ; dir.x
mov [esp+0x1C], eax    ; dir.y (unchanged)
mov [esp+0x20], ecx    ; dir.z
jmp 0x4EA125          ; normalize + integrate

```

Live cave bytes at file 0x2E5C58 (30 bytes):

```

83 7D 08 1B 75 07 8B CA BA 00 00 80 3F
89 54 24 18 89 44 24 1C 89 4C 24 20 E9 AF 44 E0 FF

```

Do not zero Z. The fan is the look they want.

Travel is +X. `dir.z` keeps the old X-spread, so the eleven crests **fan along the court**. Foam can show at both baselines (“upper and lower”) while still traveling left-to-right. That fan is part of the confirmed look. The later cave that did `xor ecx, ecx` (31 C9) killed it. See §6.

SHA-256 (verified on this box)

File	Role	SHA-256
client/FantaTennis.exe	LIVE working first cave. Use this.	c f551df8bd42f32676f8b01e54496a1b74473c90ddfa1354288f6545dea92f7c
client/FantaTennis.exe.seawave-ltr.bak	LAA-only backup (pre-cave). Site is the original 14-byte stores. Cave is zeros. VirtualSize 0x2E4C58.	eebd71a1b19eca60101a195c49999d29600e6cdd6302c4a6001fb98043f91162
evidence/.../FantaTennis.exe.ltr	DO NOT USE. This copy is the Z-spread-zero cave (31 C9).	967cc05871330521c9c7dc0ade4f9f5989fce5797d81362174c10c985ff71f3f
client/FantaTennis.exe.official	Stock official bytes (no LAA, no cave). For reference only.	5477f0827acae66976403aecdd2e9ebffeb4fa28da1fedae5f9541ec25e336c31

All four files are 3 801 088 bytes. Official vs LAA-only differ by the PE `IMAGE_FILE_LARGE_ADDRESS_AWARE` characteristic (0x20). The launch helper (`scripts/apply-ft-client-patches.py`) is an official-test stub: it **only reapplies LAA** and skips invite caves. It must not wipe the LTR cave — and the current stub does not write the cave region. An exe change still requires a **client restart**. Servers do not need a restart for the cave.

apply_ltr_patch.py / bytes.txt are stale

`evidence/seawave-ltr/apply_ltr_patch.py` and `bytes.txt` still encode the **Z-spread-zero** cave (31 C9). Re-running that script on a clean exe will install the wrong look. `first-ltr-restore.txt` is the record of putting 8B CA back. Live exe already has the first cave. Do not “fix” it with the apply script.

5. How the 11 crests look with this cave

Vanilla: `dir.x` holds the $-0.5...+0.5$ fan, `dir.z` is the travel. After normalize the wall is wide in X and walks along Y.

First cave: `swap.dir.x = +1.0` (travel LTR). `dir.z = old.dir.x` (the fan). After normalize, each crest has a slightly different Z, so the eleven pieces **fan along the court** — foam can appear toward both baselines — while the whole set still travels left-to-right. Thewind wants that. Do not collapse it.

Knockback X is still forced 0 for skill 27 at VA 0x4F1C91 (RE note; that site was not patched). Expect LTR travel with the old along-court pop. Hit numbers from the morning 9-drop run (WAVE 1 popup 9708, WAVE 4 popup 9042) were on the unpatched travel axis and are **not** re-measured for LTR tonight.

6. What we tried that looked wrong — do not use

Thewind rejected these. They are in the evidence folder so nobody “improves” the confirmed take back into them.

Attempt	What it was	Why it is wrong
Z-spread-zero cave	<code>xor ecx, ecx</code> (31 C9) so <code>dir.z = 0</code> . Same site/jmp. SHA 967cc05871330521c9c7dc0ade4f9f5989fce5797d81362174c10c985ff71f3f. On disk as <code>FantaTennis.exe.ltr</code> and in <code>apply_ltr_patch.py</code> .	Thewind 17:15–17:28 PT: foam no longer looked like the first tape. Restored 31 C9 → 8B CA at 17:29 PT; live sha back to cf551df8...
Spawn x = +200	“Player position is right” was misread as “put the drop on the right sideline.” Tapes <code>ltr-player-right*.mp4</code> (16:52–17:09 PT).	They meant the first take was already correct (wave starts on the left). +200 is the wrong side.
Spawn x = -60	“Net center a bit left.” Tapes <code>ltr-net-center*.mp4</code> (16:35–16:43 PT).	Not the look they wanted. Confirmed drop is further left: <code>x = -200</code> .
Morning “LTR is impossible”	Origin-findings PDF + first <code>ltr-wavetest.mp4</code> (13:40 PT) before the cave, or with a hook that did not change travel.	Superseded. Travel axis is client-hardcoded; the cave is what flips it.

A later `src/dist` edit at 17:44 PT set the live scripts to `x: -500` (“net line, further left (–500)”). That is **not** the confirmed take. There is an `ltr-neg500.mp4` (mtime 17:45 PT / 15:45 UTC). Thewind has not confirmed –500. To reproduce tonight’s look, put the scripts back to –200 and `-reloadScripts`.

7. Working spawn and scripts

Confirmed spawn — keep command + autopad in sync

5 waves, 5 s apart, attacker = 4, `xyz = (-200, 0, 0)`, label “net center, further left”. Packet skillId 27. Target byte in the packet is also 4 (dummy). Seed is random 0..126.

```
{ attacker: 4, x: -200, z: 0, y: 0, guess: "net center, further left" } × 5
delay = 500 + i * 5000 ms
new S2CMatchplayUseSkill(4, 4, 27, seed, -200, 0, 0)
```

Path	Role
<code>game-server/target/dist/scripts/command/wavetest.js</code>	Live command. Waiting room → writes <code>/tmp/jftse-wavetest.arm</code> , chat “Armed...”. In-match → fires at once.
<code>game-server/target/dist/scripts/event/3_wavetest_autopad.js</code>	Live event. On <code>MP_GAME_ANIM_SKIP_END</code> : pad HP, then if the arm file exists, delete it and fire the same volley.
<code>game-server/src/main/resources/scripts/command/wavetest.js</code> <code>event/3_wavetest_autopad.js</code>	Source copies. Must match dist. At write time both <code>src</code> and <code>dist</code> had been edited to <code>x: -500</code> — restore –200 before a confirmation replay.

Command behaviour (verified in the live JS)

- `-wavetest` in the waiting room = arm. Volley fires after the intro (autopad on `MP_GAME_ANIM_SKIP_END`).
- `-wavetest` in-match = immediate volley (same 5 steps).
- Guardian only. Other modes get “Only works in guardian mode.”
- `-reloadScripts` after every script edit. Events stay stale in memory otherwise. Chat should show “Reloading commands... / Reloading events... / Scripts reloaded”.
- HP pad: server `max/current 30000`, `setDead(false)`, plus client heal packet skill 1 via `S2CMatchplayDealDamage(0, 30000, 0, 1, 0.0, 0.0)`. Client HP is a **short**; 99999 overflows. The bar may still draw 200/200; the pad is what keeps a level-1 Testmon alive. Solo Testmon otherwise dies in about 15 s (origin-findings).

- These two scripts do not replace guardian-phase/10.

8. Tapes that matter

All under /workspace/jftse-work/evidence/seawave-ltr/. Mtimes below are file mtimes (UTC), shown as PT = UTC+2.

File	mtime (PT)	What it is
ltr-net-left3.mp4 (4.5 MB)	15:59 PT (13:59 UTC)	Afternoon first tape Thewind pointed at: intro, then five (−200, 0, 0) waves. This is the look.
ltr-firstcave-neg200.mp4 (25.8 MB) + -small.mp4	17:41 PT (15:41 UTC)	Tonight's restore of the first cave + −200 spawn. Confirmed 17:43 PT.
ltr-wavetest.mp4	13:40 PT (11:40 UTC)	Early LTR attempt; analysis then said "still baseline↔net". That reading of the travel axis was wrong.
ltr-net-center*.mp4	16:35–16:43 PT	x = −60 experiment. Rejected.
ltr-player-right*.mp4	16:52–17:09 PT	x = +200 experiment. Rejected.
ltr-neg200-restore*.mp4	17:18–17:23 PT	Taken while the Z-spread-zero cave was still live. Looked wrong. Do not treat as the reference.
first-ltr-restore.txt	17:29 PT (15:29 UTC)	Byte-level restore 31 C9 → 8B CA and the two SHAs.
ltr-neg500.mp4	17:45 PT (15:45 UTC)	Exists. Spawn x = −500. Not confirmed. Do not substitute for −200.

9. Ops notes

Item	Value (from files / tonight's session)
Client	FantaTennis.exe via Proton GE-Proton11-1 (proton-ge11/version).
Auth / game / relay / chat	TCP 5894 / 5895 / 5896 / 5897 (each server.conf ServerPort, plus docker-compose).
AC	Real AC on TCP 3724 (docker-compose maps 3724:3724). Need a live TCP session, not a leftover Wine process.
Splash "Initializing..."	Leftover Wine or no AC TCP. Healthy boot: AC channelActive within ~15 s, then register result:0, heartbeats ~10 s after that. Tonight 17:30 PT boot never held 3724 (8 ms probe); 17:34 PT relaunch was healthy.
Account / character	test / Testmon (Lv 01) used tonight. Reset Account.status if a leftover session blocks login. (Exact SQL is out of scope here.)
Guardian room	Lock the 3 empty slots before START enables. Mode: Guardian Battle Mode.
Skip intro	~top-right. On DISPLAY :10 at 1280×800 the working click tonight was (959, 62).
START vs Linux dock	Dock covers the bottom UI. Click the top edge of START. Tonight: ~(821, 655) on 1280×800. A low click opens the host browser.
Exe vs scripts	Cave change = restart the client. Script change = -reloadScripts, no client restart. Do not let the launch helper replace the exe with official/LAA-only.

DISPLAY :10 / 1280×800 and the two click points above are from tonight's computer-use session, not a general client constant. On another machine, find START's top edge and the intro Skip control by eye.

10. Git and related docs

Repo	https://github.com/ThewindMom/JFTSE.git
Branch with harness + morning PDF	test/wave-origin-directions — commits 2722908e (wavetest), 9319120e (findings), e450c447 (ignore SeaWave hits on guardians). Based on development 568fc3ec (origin-findings).
This PDF	LTR-cave confirmation addendum. Do not commit or push from this agent / this note.
Morning findings (superseded on travel)	JFTSE-wavetest/docs/seawave-origin-findings-2026-08-18.md + .pdf. Court origin, 9-drop table, "xyz plants the drop" still stand. "LTR is not possible" does not.
This box's JFTSE checkout	Currently on branch batt lemon (unrelated). The wavetest checkout is the origin-directions branch. Live scripts that actually fire are the dist copies under game-server/target/dist/scripts/.

Unknown / not re-verified tonight

- Fielditem_INI3.set was not decrypted. Data-clone is still a dead end for travel (RE note).
- LTR knockback feel was not re-measured. The 0x4F1C91 X-zero is still in the exe.
- Dummy-4 world position was not dumped. The cave hardcodes +1.0, so attacker Z no longer aims the wave.
- Whether $x = -500$ is “more left but same look” is unknown. Not confirmed.
- ShotSpeed / per-crest + $\text{index} * 35$ and dest-Y force (VA 0x4EA14F) were not re-read in a debugger tonight; they come from the morning RE note.
- Which Proton prefix is the one Proton will load if someone launches from a different helper is environment-specific. Confirm the live exe sha256 after launch.

Quick reference (confirmed take only)

Copy this	Value
Live exe sha256	cf551df8bd42f32676f8b01e54496a1b74473c90ddfa1354288f6545dea92f7c
LAA-only backup sha256	eebd71a1b19eca60101a195c49999d29600e6cdd6302c4a6001fb98043f91162
Z-zero (wrong) sha256	967cc05871330521c9c7dc0ade4f9f5989fce5797d81362174c10c985ff71f3f
Site file 0x0EA115	E9 3E BB 1F 00 90x9 → jmp 0x6E5C58
Cave file 0x2E5C58	83 7D 08 1B 75 07 8B CA BA 00 00 80 3F 89 54 24 18 89 44 24 1C 89 4C 24 20 E9 AF 44 E0 FF
The two bytes that decide the look	cave+6 = 8B CA (mov ecx, edx). 31 C9 is the rejected Z-zero.
RTL if ever needed	BA 00 00 80 BF (−1.0) instead of BA 00 00 80 3F (+1.0). Not tonight's take.
Spawn	5 × attacker=4 xyz=(−200, 0, 0) “net center, further left” skillId 27 5 s apart
Scripts	.../dist/scripts/command/wavetest.js and .../dist/scripts/event/3_wavetest_autopad.js — keep in sync, then -reloadScripts
.text VirtualSize	0x2E4C58 → 0x2E4C80 at file 0x220

How to reproduce tonight's take

One page. Goal: the same foam as ltr-net-left3.mp4 / ltr-firstcave-neg200.mp4. If any box fails, stop and fix that box — do not “tune” spawn or cave.

The look you are matching

Vertical foam band, traveling left → right, crests fanning toward both baselines, drop at net-center further left. Five labeled waves. Not a thin net-line. Not a right-sideline start.

■	1. Exe sha256sum client/FantaTennis.exe must be cf551df8bd42f32676f8b01e54496a1b74473c90ddfa1354288f6545dea92f7c. File 0x2E5C58 must contain 8B CA at cave+6, not 31 C9. File 0x0EA115 must be E9 3E BB 1F 00 + nops. If you just wrote the exe, restart the client. Launch helper may only set LAA.
■	2. Scripts dist command + dist autopad both have five steps { attacker: 4, x: -200, z: 0, y: 0, guess: "net center, further left" }. If they say -500 or +200 or -60, edit both files to -200. Keep them identical.
■	3. Reload In the guardian waiting room type -reloadScripts. Wait for “Scripts reloaded”. Skipping this fires yesterday’s steps from memory.
■	4. Arm Type -wavetest. Chat must say “Armed. START a guardian match; volley fires after the intro.” If you are already in a match it fires immediately — that is also fine.
■	5. Room Guardian Battle Mode. Lock the three empty slots. START is otherwise grey. Click the top edge of START (dock covers the bottom; ~821,655 on the 1280×800 :10 session).
■	6. Intro Skip ~top-right. Autopad chat: “HP padded to 30000 at match start.” Then “Wave origin test: 5 SeaWaves, 5s apart.”
■	7. Labels Each wave: WAVE n/5 attacker=4 xyz=(-200, 0, 0) net center, further left. If the numbers are not -200, you did not reload, or dist is not the file the server loaded.
■	8. Watch travel Band should be vertical and walk left → right. Crests may show “upper and lower” (both baselines). If it is a thin line on the net, you have the Z-zero cave. If it walks toward you from the far baseline as a sideline-spanning wall, you have vanilla (no cave or cave not loaded).
■	9. Record Start the recorder before START. The 15:59 PT first tape and the 17:41 PT restore both did this. A late recorder is how we lost earlier takes.
■	10. If splash sticks on Initializing Kill leftover Wine/client PIDs only (not the Java servers). Confirm something is listening on 3724 and a new boot gets AC channelActive within ~15 s + register result:0. Then retry.

If the foam matches the first tape, stop. Do not then change X, zero Z, or swap +1.0 for -1.0 “just to see.” That is how tonight’s look was lost for an hour.

Written 18 August 2026 after Thewind's 17:43 PT confirmation. Sources: live exe + backup hashes, dist/src wavetest scripts, evidence/seawave-ltr/{README.md, bytes.txt, first-ltr-restore.txt}, JFTSE-wavetest/docs/seawave-origin-findings-2026-08-18.md, server.conf ports, proton-ge11/version, apply-ft-client-patches.py, transcript b6352f6f-e679-43e5-aca3-e41ddf5fe74a.